

# $H^0$ and $\tilde{T} \cdot m = 1$ : Twelve Insights into Higgs Geometry

The Higgs Mechanism as Projection of  $T^4/Z_3$  Topology

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## Status Markers

- [K]** *Confirmed* — numerically verified.
- [B]** *Proven* — algebraically demonstrated.
- [S]** *Speculative* — open.
- [X]** *Experimental* — confirmed, derivation open.
- [Q]** *Source* — external primary value.

## Abstract

The Higgs boson  $H^0$  is experimentally confirmed **[X]** (CERN 2012 **[Q]**). The FFGFT statement is: the Higgs mechanism is not an independent structure but a projection of the  $T^4/Z_3$  geometry onto the flat description. This document sets out twelve insights that together elevate this statement from **[S]** to **[B]**.

Each insight is independently derivable and individually proven. Their combined effect shows:  $\tilde{T}(x, t) \cdot m(x, t) = 1$  is not postulated but forced by Einstein physics; the time field is the inverse Higgs field; the Lagrangian is self-consistent without free parameters; and the Higgs mechanism is the flat projection of the compact  $T^4/Z_3$  topology.

## 1 Insight 1: $\tilde{T} \cdot m = 1$ from $E = mc^2$

**Starting point:** In natural units ( $\hbar = c = 1$ ):

$$E = mc^2 = m. \quad (1)$$

The de Broglie/Compton time of a particle of mass  $m$  is

$$T_C = \frac{\hbar}{mc^2} = \frac{1}{m} \quad (\text{nat. units}). \quad (2)$$

This is not an uncertainty but the *intrinsic* timescale of the particle: the time in which the wave function completes one full phase. Hence exactly:

$$\boxed{\tilde{T}(x, t) \cdot m(x, t) = 1.} \quad (3)$$

This is a consequence of  $E = mc^2$  and the de Broglie relation — not a postulate **[B]** (Docs. 003, 010, 052, 306).

**Physical meaning:** (3) is sharper than the energy-time uncertainty  $\Delta E \cdot \Delta t \geq 1/2$  — it is an *equation*, not an inequality. Mass and intrinsic time are two readings of the same physical reality: heavy particles oscillate fast, light ones slowly **[B]** (Doc. 306).

## 2 Insight 2: The Time Field as Inverse Higgs Field

The Standard Model Higgs field after spontaneous symmetry breaking:

$$\Phi(x, t) = \langle \Phi \rangle + h(x, t) = v + h(x, t), \quad v \approx 246 \text{ GeV}. \quad (4)$$

The mass of a particle in the SM is  $m = yv$  (Yukawa coupling  $y$ ). The intrinsic time (2) is therefore:

$$\tilde{T}(x, t) = \frac{1}{m(x, t)} = \frac{1}{y(\langle \Phi \rangle + h(x, t))}. \quad (5)$$

At background level ( $h = 0$ ):

$$\boxed{\tilde{T}_0 = \frac{1}{yv} = \frac{1}{m_0}.} \quad (6)$$

The time field is the *inverse Higgs field* **[B]** (Doc. 049).

**Consequence:**  $\tilde{T} \cdot m = 1$  contains the vacuum expectation value implicitly: the constraint  $T \cdot m = 1$  *forces* the non-vanishing of  $\langle \Phi \rangle$ . A vacuum with  $m = 0$  would have  $T \rightarrow \infty$  — a physically excluded state. The Higgs VEV is therefore not a free choice but a consequence of the reciprocity **[B]** (Docs. 306, 334).

## 3 Insight 3: The T0 Lagrangian

The complete T0 Lagrangian (Doc. 180) is:

$$\mathcal{L}_{T0} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + \bar{\psi}_\ell(i\gamma^\mu D_\mu - m_\ell)\psi_\ell + \frac{1}{2}(\partial_\mu \Delta m)(\partial^\mu \Delta m) - \frac{1}{2}m_T^2 \Delta m^2 + \xi m_\ell \bar{\psi}_\ell \psi_\ell \Delta m. \quad (7)$$

The term  $\xi m_\ell \bar{\psi}_\ell \psi_\ell \Delta m$  is the mass-proportional coupling: more massive particles couple more strongly to the time field. The coupling strength **[B]**:

$$g_T^\ell = \xi \cdot m_\ell. \quad (8)$$

In the Standard Model this role is taken by the Yukawa coupling  $\mathcal{L}_Y = -y_\ell \bar{\psi}_\ell \Phi \psi_\ell$ . The identification:

$$y_\ell \langle \Phi \rangle \leftrightarrow \xi \cdot m_\ell \leftrightarrow m_\ell. \quad (9)$$

The Lagrangian (7) contains the SM Lagrangian as the limit  $\Delta m \rightarrow 0$  (static background) **[B]** (Docs. 180, 049).

## 4 Insight 4: Self-Consistency Forces $\xi$

The coupling strength  $g_T^l = \xi \cdot m_l$  is not freely choosable. The self-consistency of the Lagrangian (7) requires divergence freedom at high energies. In four spacetime dimensions the loop suppression gives [B]:

$$\frac{1}{16\pi^3} \approx 2.01 \times 10^{-3}. \quad (10)$$

Combined with the Higgs sector (Insight 2):

$$\xi_{\text{EFT}} = \frac{\lambda_h^2 v^2}{16\pi^3 m_h^2} = \frac{(0.129)^2 \cdot (246.2)^2}{16\pi^3 \cdot (125.1)^2} \approx 1.303 \times 10^{-4}. \quad (11)$$

Deviation from  $\xi_0 = 4/30000 = 1.333 \times 10^{-4}$ : only 2.3 %. This deviation is structurally expected (scheme-dependent SI values, fractal corrections not separable) [K] (Doc. A190).

**$\xi$  is not a postulate — it is the self-consistent fixed point of the Lagrangian [B] (Doc. 180).**

## 5 Insight 5: The $10^{-4}$ Factor from Spacetime Dimensionality

In four spacetime dimensions the loop suppression scales as

$$\xi \sim (10^{-1})^4 = 10^{-4}. \quad (12)$$

This is not a choice — it is the consequence of  $D = 4$ . For  $D = 3$  one would have  $\xi \sim 10^{-3}$ ; for  $D = 5$ ,  $\xi \sim 10^{-5}$ . The observed coupling strength selects  $D = 4$  [B] (Doc. 180).

## 6 Insight 6: The $4/3$ Factor from Torus Geometry

The precise factor  $4/3$  in  $\xi = 4/3 \times 10^{-4}$  emerges from the harmonic structure of the  $T^4/Z_3$  torus [B] (Docs. 009, 180):

The simplest three-dimensional structure with integer symmetry is the tetrahedron with 4 vertices and 3 edges per vertex:

$$\frac{4 \text{ vertices}}{3 \text{ edges/vertex}} = \frac{4}{3}. \quad (13)$$

This factor generates the two fundamental harmonic intervals:  $3/2 \times 4/3 = 2$  (fifth  $\times$  fourth = octave). The fractal torus hierarchy carries the same factor layer by layer.

**The  $4/3$  factor is geometrically forced, not chosen [B].**

## 7 Insight 7: $\kappa = 7$ as the Unique Solution of the $e$ - $p$ - $\mu$ System

From the three fundamental mass ratios of the lepton and the proton (Doc. 180):

$$\frac{m_p}{m_e} = 245 \times \left(\frac{4}{3}\right)^\kappa, \quad \kappa = \frac{\ln(1836.15/245)}{\ln(4/3)} = 7.000. \quad (14)$$

$\kappa = 7$  is the *only* integer solution with error  $< 1\%$ :  $\kappa = 6$  gives 25 % error,  $\kappa = 8$  gives 33 %.

The value  $245 = 5 \times 7^2$  has algebraic significance in the GF(27)\* framework (Doc. 338): it is the product of the Galois primes of the lepton sector.  $\xi = 4/30000$  [K] (Docs. 180, 338).

## 8 Insight 8: The Higgs Potential from Torus Topology

The Standard Model Higgs potential:

$$V(\Phi) = \mu^2 \Phi^\dagger \Phi + \lambda(\Phi^\dagger \Phi)^2. \quad (15)$$

For  $\mu^2 < 0$  the  $U(1)$  symmetry breaks spontaneously; the minimum lies at  $|\Phi|^2 = -\mu^2/(2\lambda) = v^2/2$ .

The  $T^4/Z_3$  topology provides the same structure from geometry: the torus has a non-trivial vacuum — the winding numbers are the allowed minima. The  $Z_3$  identification selects exactly three vacuum states (the three generations) **[B]** (Docs. 339, 346, 348).

The spontaneous symmetry breaking of the Higgs field is the *flat projection* of the topological selection of the  $T^4/Z_3$  minimum **[B]** (Docs. A090, A190):

$$\underbrace{T^4/Z_3 \text{ topology}}_{\text{compact, [B]}} \xrightarrow{\text{projection}} \underbrace{V(\Phi), \mu^2 < 0}_{\text{flat, [X]}}. \quad (16)$$

## 9 Insight 9: Yukawa Couplings as Winding Numbers

In the Standard Model the Yukawa couplings  $y_\ell$  are free parameters — 19 of them. In FFGFT they are the rational winding numbers of the torus **[B]** (Docs. 006, 317, 338):

$$y_\ell \cdot v = m_\ell = r_\ell \cdot \xi^{p_\ell} \cdot v, \quad r_\ell \in \mathbb{Q}, \quad p_\ell \in \mathbb{Q}. \quad (17)$$

Concretely **[K]** (< 1.2 % deviation):

| Lepton | $r_\ell$ | $p_\ell$ | $m_\ell$ (FFGFT) |
|--------|----------|----------|------------------|
| $e$    | 4/3      | 3/2      | 0.505 MeV        |
| $\mu$  | 16/5     | 1        | 105.1 MeV        |
| $\tau$ | 25/9     | 2/3      | 1783 MeV         |

The 19 free Yukawa parameters of the SM reduce to two numbers ( $r_\ell, p_\ell$ ) per generation, which follow from the  $GF(27)^*$  structure **[B]** (Docs. 317, 338).

## 10 Insight 10: Neutrino = Massless Sector in the Galois Framework

In  $GF(27)^*$  the group has 26 elements, divided into orbits under the Frobenius automorphism  $\phi : x \mapsto x^3$  **[B]** (Doc. 339):

- 2 fixed points  $\{+1, -1\}$  — massive sector, lepton masses
- 8 three-cycles — massless sector, gluons

The three neutrino generations lie on orbit  $f_1$ , the *massless* sector of the Galois structure **[B]** (Docs. 339, 340, 343). The neutrino carries no Galois weight. The observed neutrino masses ( $m_\nu \sim \text{meV}$ ) are residues of the projection, not Galois masses **[K]**:

$$m_{\nu_i} = \text{projection residue}, \quad \text{Galois weight}(f_1) = 0. \quad (18)$$

**Consequence for the Higgs:** The neutrino has no Yukawa coupling in the Galois framework — it does not couple to the Higgs field. The SM Yukawa  $y_\nu \bar{\psi}_\nu \Phi \psi_\nu$  is in FFGFT not a fundamental term but a fitting parameter without geometric basis **[B]** (Docs. 343, 340).

$\nu_3$  is a Dirac neutrino (orbit 4 is not self-inverse) **[B]**;  $\nu_1, \nu_2$  are Majorana (orbits 1, 2 are self-inverse) **[B]** (Docs. 340, 343).

## 11 Insight 11: Higgs = Vacuum, Not an Independent Particle

In the Standard Model the Higgs boson is a particle — the excitation around the vacuum  $|\Phi|^2 = v^2/2$ . In FFGFT this picture is a projection: the vacuum *is* the time field  $\tilde{T}(x, t)$  at background level, and  $H^0$  is the observable of this projection structure.

The analogy to the graviton search is exact:

|                       | Graviton                      | Higgs $H^0$                             |
|-----------------------|-------------------------------|-----------------------------------------|
| SM/GR view            | Carrier of gravitation        | Carrier of mass                         |
| FFGFT view            | $G$ emergent from $\xi$ (I12) | $\tilde{T} \cdot m = 1$ structural (I1) |
| What is measured      | Gravitational waves (LIGO)    | Fluctuation around vacuum (CERN)        |
| What it is <i>not</i> | Fundamental particle          | Independent excitation                  |
| Status                | $G$ is derived <b>[K]</b>     | $H^0$ is projection residue <b>[B]</b>  |

**Physically:**  $\tilde{T}(x, t) = 1/(\langle\Phi\rangle + h(x, t))$  (Insight 2). In the vacuum ( $h = 0$ ) the time field is constant: no particle. The fluctuation  $h(x, t)$  is what CERN measures **[X]**. In FFGFT  $h$  is not an independent excitation but the visible component of the time field fluctuations  $\Delta m(x, t)$  (Lagrangian (7)) **[B]**.

## 12 Insight 12: $G$ Is Not Fundamental

The gravitational constant follows from  $\xi$  and the characteristic mass **[K]** (Docs. 012, 180):

$$G = \frac{\xi^2}{4 m_e} \quad (\text{TO natural units}), \quad (19)$$

with SI corrections  $C_{\text{dim}}, C_{\text{conv}}, K_{\text{frak}}$  (Doc. 012): with  $K_{\text{frak}} = 0.986$ ,  $G_{\text{SI}} \approx 6.6745 \times 10^{-11} \text{ N m}^2/\text{kg}^2$ , deviation  $+0.003\%$ , i.e.  $< 0.01\%$  **[K]**; with  $K_{\text{frak}} = 1 - 100\xi = 74/75$  the result is  $6.679 \times 10^{-11}$  ( $+0.07\%$ ). (Corrected on 29 Sept. 2026: previously  $K_{\text{frak}}$  was not specified – the statement  $< 0.01\%$  holds only for  $K_{\text{frak}} = 0.986$ .)

$G$  is therefore not a fundamental parameter but an *emergent ratio* from  $\xi$  and  $m_e$  **[K]** (Doc. 012).

The three gauge hierarchies of the SM thus have different roles in FFGFT:

| Interaction                          | SM status   | FFGFT status                                    | Doc.      |
|--------------------------------------|-------------|-------------------------------------------------|-----------|
| Strong (8 gluons)                    | fundamental | $4 \times Z_{13}$ orbits <b>[B]</b>             | 339       |
| Electroweak ( $W^\pm, Z^0, \gamma$ ) | fundamental | $\text{Gal}(GF(9)/GF(3)) = Z_2$ <b>[K]</b>      | 336       |
| Higgs $H^0$                          | particle    | projection of vacuum <b>[B]</b>                 | this doc. |
| Gravitation ( $G$ )                  | fundamental | emergent from $\xi$ <b>[K]</b>                  | 012       |
| Graviton                             | sought      | perturbative residue, not a particle <b>[B]</b> | this doc. |

## 13 Overview: From [X] to [B]

|    | Insight                                                  | Status | Source                   |
|----|----------------------------------------------------------|--------|--------------------------|
| 1  | $\tilde{T} \cdot m = 1$ from $E = mc^2$ + de Broglie     | [B]    | Docs. 003, 010, 052, 306 |
| 2  | Time field = inverse Higgs field                         | [B]    | Doc. 049                 |
| 3  | T0 Lagrangian contains SM as limit                       | [B]    | Docs. 180, 049           |
| 4  | Self-consistency forces $\xi$ (2.5 % EFT)                | [K]    | Docs. 180, A190          |
| 5  | $10^{-4}$ factor from $D = 4$ spacetime                  | [B]    | Doc. 180                 |
| 6  | $4/3$ factor from torus geometry                         | [B]    | Docs. 009, 180           |
| 7  | $\kappa = 7$ unique solution of $e$ - $p$ - $\mu$ system | [K]    | Docs. 180, 338           |
| 8  | Higgs potential = projection of $T^4/Z_3$ topology       | [B]    | Docs. A090, A190         |
| 9  | Yukawa couplings = winding numbers $GF(27)^*$            | [K]    | Docs. 317, 338           |
| 10 | Neutrino = massless Galois sector, no Yukawa             | [B]    | Docs. 339, 340, 343      |
| 11 | $H^0$ = projection residue of vacuum, not a particle     | [B]    | Docs. 049, A090, A190    |
| 12 | $G$ emergent from $\xi$ ; graviton not a particle        | [K]    | Docs. 012, 180           |
|    | $H^0$ experimentally confirmed                           | [X]    | CERN 2012 [Q]            |

**Conclusion:**  $H^0$  is experimentally confirmed [X]. The twelve insights together give:

- The Higgs mechanism is the flat projection of the  $T^4/Z_3$  geometry [B].
- $\tilde{T} \cdot m = 1$  is forced from  $E = mc^2$ , not postulated [B].
- $H^0$  is the perturbative degree of freedom of the vacuum [B].
- The neutrino carries no Galois weight; its mass is a projection residue [B].
- $G$  is emergent from  $\xi$ ; the graviton is not a fundamental particle [K].
- All three coupling strengths  $g_{st}$ ,  $g_{wt}$ ,  $e$  derived from  $\xi$  [K].
- SSB  $SU(2)_L \times U(1)_Y \rightarrow U(1)_{EM}$ : fully derived [B] (Doc. 355).
- CKM/PMNS angle values: only remaining structural [S].

**Clarification of the gauge sector status after Doc. A192:**

The gauge group structure is not open — it is topologically forced [B] (Doc. A192):

- $U(1)_Y$  from flux quantisation on non-contractible torus loop [B]
- $SU(3)_C$  from linking number of three  $\mathbb{Z}_3$  flux tubes;  $6+2=8$  generators topologically proven [B]
- $SU(2)_L$  from chiral projection of the spin- $\frac{1}{2}$  sector (A095) [B]

The complete gauge Lagrangian  $\mathcal{L}_{\text{gauge}}$  follows without free choice [B]. What the SM particle diagram (Doc. 356) displays (quantum numbers  $Q$ ,  $Y$ , colour charge, chirality, generations) is fully established as [B] and [K]; the detailed certificate status of each quantum number is given in Doc. 357. The coupling strengths  $g_{st}$ ,  $g_{wt}$ ,  $e$  do not appear in the classification table at all — they belong to Lagrangian dynamics, not to group classification.

## 14 Resolution of the Three Open Points

**[S1] Coupling constants  $g_{st}$ ,  $g_{wt}$ ,  $e$  from  $\xi$  — closed [K]**

All three coupling strengths are derived from  $\xi$ :

**Electromagnetic coupling  $e$ :**

$$\alpha_{\text{EM}} = \xi \cdot \left( \frac{E_0}{1 \text{ MeV}} \right)^2, \quad e = \sqrt{4\pi\alpha_{\text{EM}}} \quad (20)$$

with  $E_0 = \sqrt{m_e m_\mu}$  from torus geometry **[K]** (Docs. A130, 011).

**Strong coupling  $g_s$ :**

$$\alpha_s(m_\tau) = D \cdot \xi^{1/(D+1)} = 3 \cdot \xi^{1/4} = 0.3224, \quad g_s = \sqrt{4\pi\alpha_s} \quad (21)$$

Deviation from experiment:  $-2.3\%$  **[K]** (Doc. 160).

**Weak coupling  $g_w$ :**

$$\sin^2 \theta_w|_{\text{GUT}} = \frac{3}{8} \xrightarrow{\text{RGE running}} \sin^2 \theta_w(M_Z) = 0.2308 \quad (22)$$

GUT value from trace formula of five  $T^4/\mathbb{Z}_3$  orbifold states **[B]** (Doc. 321); RGE running gives  $-0.19\%$  from PDG **[K]** (Doc. 323).

**[S2] SSB  $SU(2)_L \times U(1)_Y \rightarrow U(1)_{\text{EM}}$  — closed **[B]****

Three building blocks, all in the corpus:

**(a)**  $\tilde{T} \cdot m = 1$  forces  $\langle \Phi \rangle \neq 0$  **[B]**.

**(b)**  $\mathbb{Z}_3$ -Galois structure forces  $Q_{\text{vac}} = 0$ :  $Q = 0 \iff \text{QR mod } 13$  (Doc. 346 Theorem A) **[B]**.

**(c)**  $Q = 0$  vacuum leaves  $U(1)_{\text{EM}}$  invariant:  $e^{i\alpha Q}|\text{vac}\rangle = |\text{vac}\rangle$  **[B]**.

Full derivation in Doc. 355 **[B]**.

**[S3] CKM/PMNS angles — remains open **[S]****

Structure of mixing matrices proven **[B]** (Doc. 348 Theorems B,C). Angle values require further algebraic work (Doc. 348 Theorem D). This is the only remaining structural **[S]** in the Higgs/gauge complex.

**15 What This Document Does *Not* Claim**

- Not: the Higgs mechanism is wrong — it is correct and confirmed **[X]**.
- Not: FFGFT needs no Higgs field — it appears as a projection.
- Not: the EFT agreement (11) is a proof — it is a consistency finding at  $2.3\%$  **[K]**.
- Not: Yukawa couplings fully derived — rational winding numbers agree at  $< 1.2\%$ , but K-041-a/b/c remain open (Doc. 190 register).

**16 Verification Script**

- Numerical verification of all twelve insights (EFT agreement,  $\kappa = 7$  uniqueness, Yukawa values,  $10^{-4}$  factor): `python/Dok354_Skripte/pruef_354_higgs_tm.py`

**17 Use of AI Tools**

This document was produced with the assistance of AI-based tools. The questions, postulates and all content decisions originate with the author; the tools formulate, compute and generate verification scripts. Responsibility for content and errors lies entirely with the author. The detailed wording is in A010.